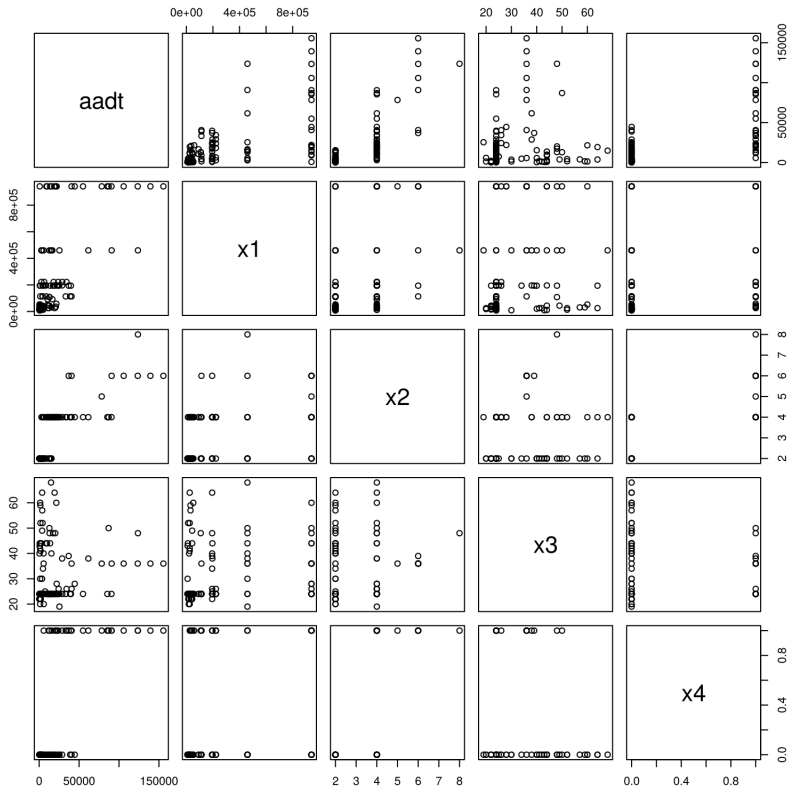
Modelling Average Annual Daily Traffic (AADT): A Multiple Regression Analysis

Introduction

The data set aadt.txt has 121 road sections. The response is the average annual daily traffic (AADT), the average number of vehicles passing through a section of road each day over a year. The four predictors are X_1, the population of the county the road section is in; X_2, the number of lanes; X_3, the width of the road section in feet; and X_4, whether there is control of access to the road section. In the data file X_4 is coded 1 for access control and 2 for no access control, so we re-coded it as a dummy variable with $X_4 = 1$ for access control (27 sections) and $X_4 = 0$ for no access control (94 sections). We follow the procedure in the course notes: graphical display of the data, fitting a model, checking its adequacy, simplifying the model, and making inference.

Graphical Display of the Observed Data

To understand the initial relationships between the variables, a scatter plot matrix was generated for the first five columns of the dataset.



Explanation:

- AADT increases with the number of lanes (X_2) and is much higher on roads with access control (X_4 = 1). It also tends to rise with county population (X_1), although the spread is large. Road width (X_3) shows no clear relationship with AADT.
- AADT is strongly right-skewed: most sections carry fewer than 25,000 vehicles a day, while a few carry over 100,000. The spread of AADT grows as X_1 and X_2 increase, which is an early sign of non-constant variance.
- X_2 and X_4 are related (sample correlation 0.66), since roads with access control usually have four or more lanes. X_1 takes only a few distinct values because many road sections lie in the same county.

Fitting the Initial Model

A multiple linear regression model was initially fitted using all four predictor variables.

```
> summary(mlr)

Call:
lm(formula = aadt ~ x1 + x2 + x3 + x4, data = aadt_data)

Residuals:
```

```
   Min     1Q Median     3Q    Max
-36263  -8501   3493   6018  68317

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.604e+04  5.255e+03  -4.955 2.49e-06 ***
x1           3.303e-02  4.708e-03   7.017 1.63e-10 ***
x2           9.158e+03  1.531e+03   5.983 2.49e-08 ***
x3           1.003e+02  1.243e+02   0.807    0.421
x4           2.361e+04  4.520e+03   5.223 7.83e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

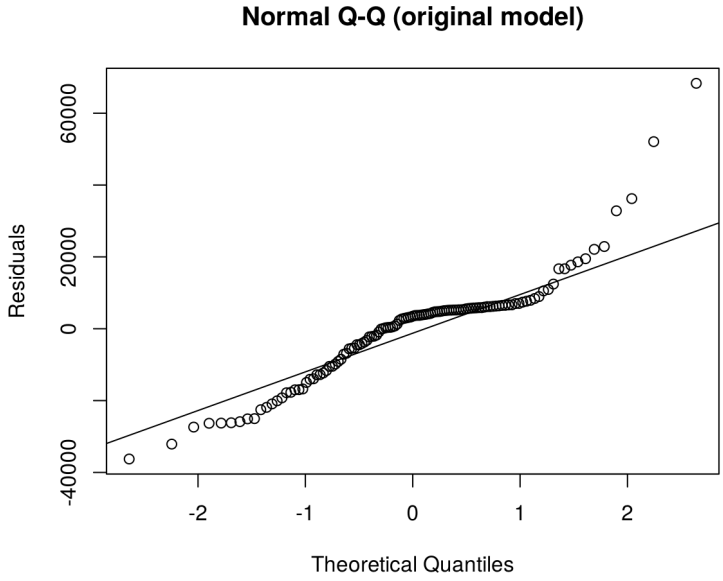
Residual standard error: 15290 on 116 degrees of freedom
Multiple R-squared:  0.7527,  Adjusted R-squared:  0.7442
F-statistic: 88.29 on 4 and 116 DF,  p-value: < 2.2e-16
```

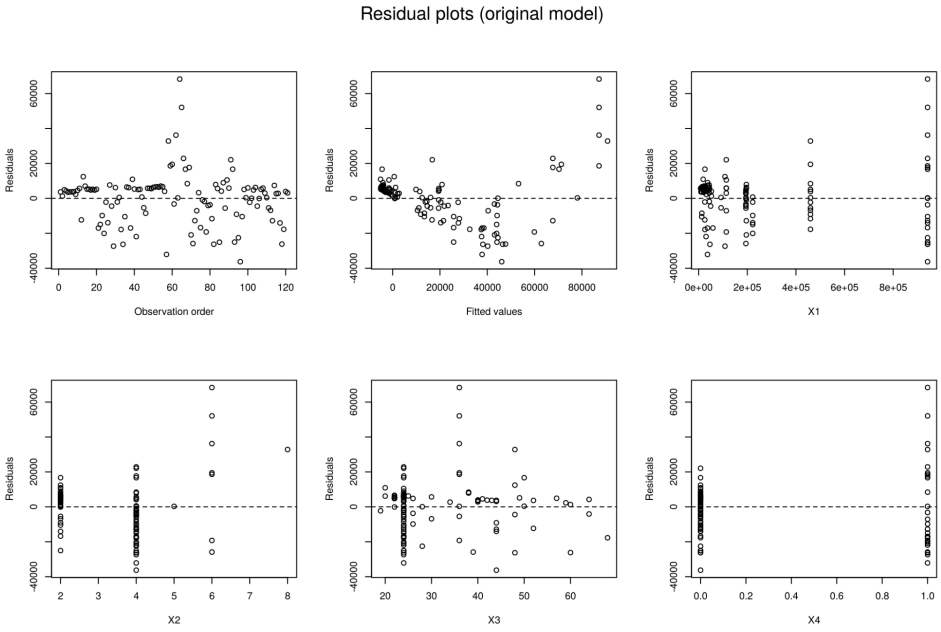
Explanation of the Fitted Model:

- **Significance:** Based on the t-tests (Pr(>|t|) column), X_1, X_2 and X_4 are significant at the 0.1% level, while X_3 (road width) is not significant (p = 0.421).
- **R-squared:** Multiple R-squared is 0.7527 (adjusted R-squared 0.7442), so the model explains about 75% of the variation in AADT. The F-statistic of 88.29 on 4 and 116 degrees of freedom (p < 2.2e-16) shows that the regression as a whole is significant.
- **Fitted model:** AADT = −26036 + 0.03303 X_1 + 9158 X_2 + 100.3 X_3 + 23610 X_4. Holding the other predictors fixed, each extra lane adds about 9,160 vehicles a day, and access control adds about 23,600.
- **Problem:** 45 of the 121 fitted values are negative (the smallest is about −5,100), which is impossible for a traffic count. This is a first sign that a linear model on the original scale does not suit these data.

Model Adequacy Checking

To check if the fitted model is adequate, we analyzed the residuals for normality, constant variance, and sequential dependence.






```
> dwtest(aadt ~ x1 + x2 + x3 + x4, data=aadt_data)

  Durbin-Watson test

data:  aadt ~ x1 + x2 + x3 + x4
DW = 1.3137, p-value = 3.101e-05
alternative hypothesis: true autocorrelation is greater than 0
```

Explanation:

- **Normality:** The points bend away from the line at both ends. The upper tail is long, with a few very large positive residuals (up to about 68,000), so the residuals are not normally distributed.
- **Variance:** The residuals against fitted values do not form a horizontal band. They curve (positive at small fitted values, mostly negative between about 30,000 and 50,000, and large and positive above 60,000), and their spread increases with the fitted value. The plots against X_1, X_2 and X_4 also fan out at larger values, so the variance is not constant.
- **Sequential dependence:** The Durbin-Watson statistic is d = 1.314 with p = 3.1e-05, which suggests positive serial correlation. However, the road sections have no time order, so this mostly reflects how the rows are arranged in the file. Sections from the same county and of similar road type are listed next to each other, which shows up as runs in the residuals against observation order.
- Because of these violations, the initial model is not fully adequate, requiring a transformation.

Model Transformation

To correct the non-constant variance observed above, a log transformation was applied to the response variable, resulting in the model: log(AADT) ~ X_1 + X_2 + X_3 + X_4. The log is a natural choice here because AADT is positive and strongly right-skewed, and the course notes suggest transforming the response when the spread grows with its level.

```
> summary(mlr_log)

Call:
lm(formula = log(aadt) ~ x1 + x2 + x3 + x4, data = aadt_data)

Residuals:
     Min       1Q   Median       3Q      Max
-2.31522 -0.46112  0.00963  0.59348  1.86778

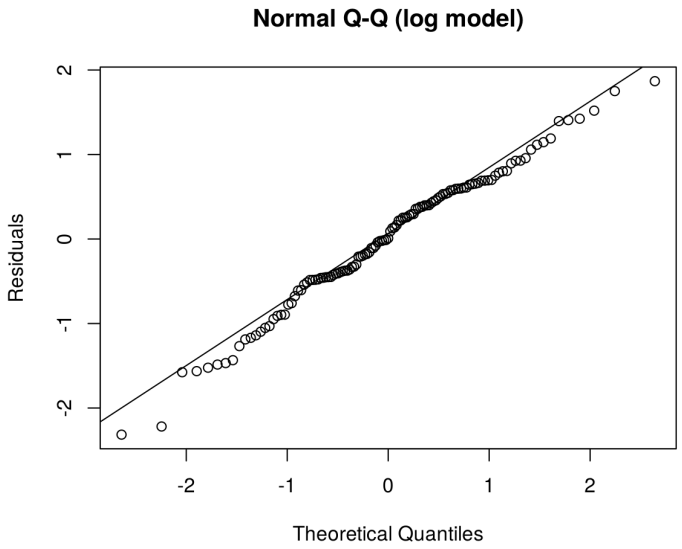
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.074e+00  2.827e-01  21.486  < 2e-16 ***
x1          1.528e-06  2.533e-07   6.034 1.96e-08 ***
x2          6.852e-01  8.234e-02   8.321 1.93e-13 ***
x3          2.501e-03  6.686e-03   0.374   0.7091
x4          6.145e-01  2.432e-01   2.527   0.0129 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

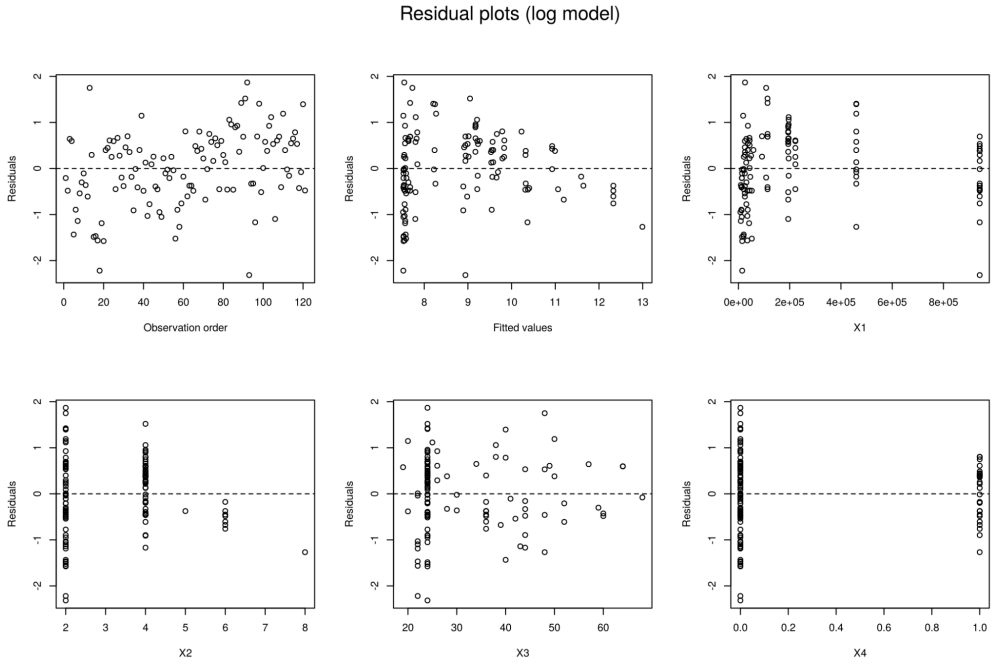
Residual standard error: 0.8227 on 116 degrees of freedom
Multiple R-squared:  0.7411,  Adjusted R-squared:  0.7321
F-statistic: 82.99 on 4 and 116 DF,  p-value: < 2.2e-16

> dwtest(log(aadt) ~ x1 + x2 + x3 + x4, data=aadt_data)

  Durbin-Watson test

data:  log(aadt) ~ x1 + x2 + x3 + x4
DW = 1.3708, p-value = 0.0001136
alternative hypothesis: true autocorrelation is greater than 0
```





Explanation of Improvement:

- In the log model, X_1 and X_2 are still significant at the 0.1% level and X_4 at the 5% level ($p = 0.0129$), while X_3 is still not significant ($p = 0.709$). The R-squared of 0.7411 cannot be compared directly with 0.7527 from the original model, because the response is now on a different scale.
- The residuals against fitted values now lie in a roughly horizontal band between about −2.3 and 1.9, with no funnel shape, so the log transformation improves the residual plots and has reduced the large differences in variance in the original model. There is a slight

hump in the middle, and the few roads with 6 lanes all have negative residuals, but neither pattern is strong. All fitted values are now positive once transformed back to vehicle counts.

- The points in the Q-Q plot now lie close to the line, with only a small departure in the lower tail, so the normality assumption is reasonable.
- The Durbin-Watson statistic is d = 1.371 with p = 0.0001, so there is still significant positive serial correlation. As with the original model, the road sections have no time order, and the correlation comes from neighbouring rows in the file being similar road sections. We note this as a limitation, since it means the standard errors may be somewhat too small.

Simplifying the Model

X_3 (road width) is not significant in the log model (p = 0.709), so we test whether it can be dropped, that is H_0: $\beta_3 = 0$ against H_1: $\beta_3 \neq 0$. We fit the reduced model log(AADT) ~ X_1 + X_2 + X_4 and compare it with the full log model using an F-test.

```
> summary(mlr_log2)

Call:
lm(formula = log(aadt) ~ x1 + x2 + x4, data = aadt_data)

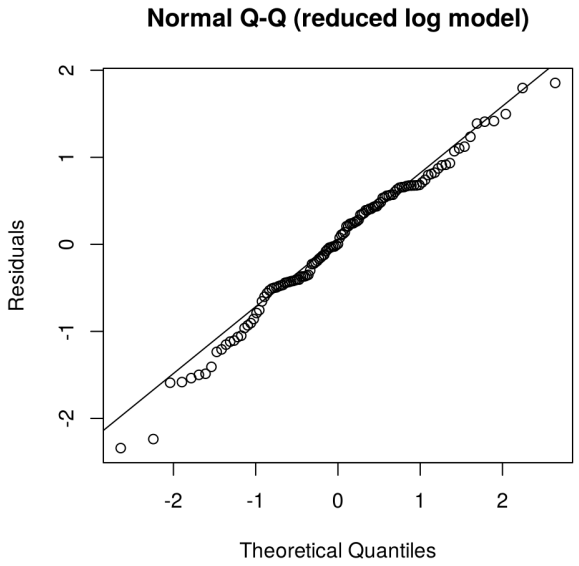
Residuals:
     Min       1Q   Median       3Q      Max
-2.34096 -0.46606  0.00488  0.56969  1.85416

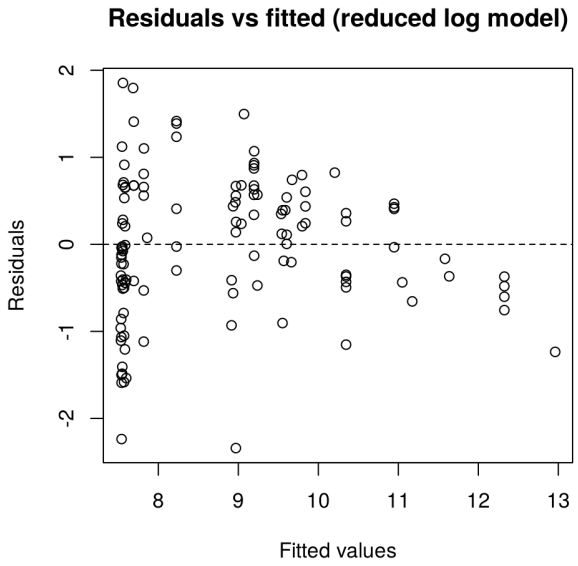
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.140e+00  2.206e-01  27.830  < 2e-16 ***
x1          1.541e-06  2.499e-07   6.170 1.01e-08 ***
x2          6.890e-01  8.142e-02   8.462 8.71e-14 ***
x4          6.005e-01  2.394e-01   2.508   0.0135 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8197 on 117 degrees of freedom
Multiple R-squared:  0.7407,  Adjusted R-squared:  0.7341
F-statistic: 111.4 on 3 and 117 DF,  p-value: < 2.2e-16

> anova(mlr_log2, mlr_log)
Analysis of Variance Table

Model 1: log(aadt) ~ x1 + x2 + x4
Model 2: log(aadt) ~ x1 + x2 + x3 + x4
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    117 78.607
2    116 78.512  1   0.09468 0.1399 0.7091
```





Explanation:

- The F-statistic is 0.14 on 1 and 116 degrees of freedom with p = 0.709, so we cannot reject H_0 at any usual significance level. Dropping X_3 barely changes the residual sum of squares (78.512 to 78.607). Because only one coefficient is tested, this F-test agrees with the t-test for X_3 in the log model ($F = t^2 = 0.374^2 \approx 0.14$).
- In the reduced model all three predictors are significant (X_1 and X_2 at the 0.1% level, X_4 at the 5% level), and the adjusted R-squared goes up slightly from 0.7321 to 0.7341. The residual plots in Figure 6 look the same as for the full log model, so removing width is an adequate choice. The F-test result (p-value = 0.709) also supports retaining the reduced model.
- The final fitted model is $\log(AADT) = 6.140 + 1.541\times10^{-6}\, X_1 + 0.6890\, X_2 + 0.6005\, X_4$.
- On the original scale, holding the other predictors fixed, each extra lane multiplies the median AADT by $\exp(0.6890) \approx 1.99$, access control multiplies it by $\exp(0.6005) \approx 1.82$, and an extra 100,000 people in the county multiplies it by $\exp(0.1541) \approx 1.17$.

Making Inference (Prediction)

Using the final reduced log model, a prediction was made for a road section with a population of 50,000 ($X_1 = 50000$), 3 lanes ($X_2 = 3$), a width of 60 feet ($X_3 = 60$), and no access control (original $X_4 = 2$, recoded to $X_4 = 0$).

Because the model predicts log(AADT), the outputs were exponentiated to return standard vehicle counts. Since X_3 is not in the final model, the road width of 60 feet does not change the prediction.

```
> predict(mlr_log2, con, interval='confidence', level=0.95)
       fit      lwr      upr
```

```
1 8.283909 8.080956 8.486861
> predict(mlr_log2, con, interval='prediction', level=0.95)
       fit      lwr      upr
1 8.283909 6.647967 9.919851
> exp(predict(mlr_log2, con, interval='confidence', level=0.95))
       fit      lwr      upr
1 3959.641 3232.322 4850.616
> exp(predict(mlr_log2, con, interval='prediction', level=0.95))
       fit      lwr      upr
1 3959.641 771.2145 20329.95
```

Results:

- **Predicted AADT:** about 3,960 vehicles per day (exp(8.2839) = 3959.6).
- **95% Confidence Interval:** [3,232, 4,851] vehicles per day.
- **95% Prediction Interval:** [771, 20,330] vehicles per day.
- The confidence interval covers the typical AADT of all road sections with these characteristics, while the prediction interval is for a single new road section. The prediction interval is much wider because it also includes the variation of an individual section around that typical value.
- The model was fitted on the log scale, so exponentiating the fit estimates the median AADT rather than the mean, and the intervals are not symmetric about the point estimate.
- No road section in the data has exactly 3 lanes (the data have 2, 4, 5, 6 and 8 lanes), so this prediction lies between the 2-lane and 4-lane roads in the data. A population of 50,000 is within the observed range of X_1.

Appendix: R Code

```
# MH3510 Group Assignment

# ---------- 1. Graphical display of the observed data ----------
aadt_raw <- read.table('aadt.txt', header=FALSE)

# only the first 5 columns are used
# x4 in the file is 1 = access control, 2 = no access control,
# so we turn it into a 0/1 dummy (1 = access control, 0 = none)
aadt_data <- data.frame(
  aadt = aadt_raw$V1,   # average annual daily traffic (response)
  x1 = aadt_raw$V2,     # county population
  x2 = aadt_raw$V3,     # number of lanes
  x3 = aadt_raw$V4,     # road width (feet)
  x4 = ifelse(aadt_raw$V5 == 1, 1, 0)
)

# scatter plot matrix of y against every predictor
plot(aadt_data)

# ---------- 2. Fit the multiple linear regression model ----------
mlr <- lm(aadt ~ x1 + x2 + x3 + x4, data=aadt_data)
summary(mlr)
```

```
# ---------- 3. Adequacy checking (original model) ----------
# normality: Q-Q plot of the residuals
qqnorm(residuals(mlr), main='Normal Q-Q (original model)', ylab='Residuals')
qqline(residuals(mlr))

# residuals against row order, fitted values and each predictor
# (fitted values / predictors go on the x-axis, residuals on the y-axis)
par(mfrow=c(2,3))
plot(residuals(mlr), ylab='Residuals', xlab='Observation order')
abline(h=0, lty=2)
plot(fitted(mlr), residuals(mlr), ylab='Residuals', xlab='Fitted values')
abline(h=0, lty=2)
plot(aadt_data$x1, residuals(mlr), ylab='Residuals', xlab='X1')
abline(h=0, lty=2)
plot(aadt_data$x2, residuals(mlr), ylab='Residuals', xlab='X2')
abline(h=0, lty=2)
plot(aadt_data$x3, residuals(mlr), ylab='Residuals', xlab='X3')
abline(h=0, lty=2)
plot(aadt_data$x4, residuals(mlr), ylab='Residuals', xlab='X4')
abline(h=0, lty=2)
par(mfrow=c(1,1))

# sequential dependence: Durbin-Watson test
library(lmtest)
dwtest(aadt ~ x1 + x2 + x3 + x4, data=aadt_data)

# ---------- 3b. Transformation of the response ----------
# the residual spread grows with the fitted values (funnel shape),
# so we try log(aadt) to stabilise the variance
mlr_log <- lm(log(aadt) ~ x1 + x2 + x3 + x4, data=aadt_data)
summary(mlr_log)

# redo the adequacy checks on the log model
qqnorm(residuals(mlr_log), main='Normal Q-Q (log model)', ylab='Residuals')
qqline(residuals(mlr_log))

par(mfrow=c(2,3))
plot(residuals(mlr_log), ylab='Residuals', xlab='Observation order')
abline(h=0, lty=2)
plot(fitted(mlr_log), residuals(mlr_log), ylab='Residuals', xlab='Fitted values')
abline(h=0, lty=2)
plot(aadt_data$x1, residuals(mlr_log), ylab='Residuals', xlab='X1')
abline(h=0, lty=2)
plot(aadt_data$x2, residuals(mlr_log), ylab='Residuals', xlab='X2')
abline(h=0, lty=2)
plot(aadt_data$x3, residuals(mlr_log), ylab='Residuals', xlab='X3')
abline(h=0, lty=2)
plot(aadt_data$x4, residuals(mlr_log), ylab='Residuals', xlab='X4')
abline(h=0, lty=2)
par(mfrow=c(1,1))

dwtest(log(aadt) ~ x1 + x2 + x3 + x4, data=aadt_data)

# ---------- 4. F-test for reduced model vs full model ----------
# x3 (road width) is not significant in the log model,
# so test H0: beta3 = 0 by comparing the model without x3 to the full one
mlr_log2 <- lm(log(aadt) ~ x1 + x2 + x4, data=aadt_data)
summary(mlr_log2)
anova(mlr_log2, mlr_log)

# quick check that the reduced model's residuals still look fine
par(mfrow=c(1,2))
```

```
qqnorm(residuals(mlr_log2), main='Normal Q-Q (reduced log model)', ylab='Residuals')
qqline(residuals(mlr_log2))
plot(fitted(mlr_log2), residuals(mlr_log2), ylab='Residuals', xlab='Fitted values',
     main='Residuals vs fitted (reduced log model)')
abline(h=0, lty=2)
par(mfrow=c(1,1))

# ---------- 5. Making inference: prediction ----------
# new road: x1 = 50000, x2 = 3, x3 = 60, x4 = 2 in the original coding (no access
#control),
# which is x4 = 0 after our recoding. x3 is kept in the data frame but the reduced
#model ignores it
con <- data.frame(x1=50000, x2=3, x3=60, x4=0)

# predictions come out on the log scale
predict(mlr_log2, con, interval='confidence', level=0.95)
predict(mlr_log2, con, interval='prediction', level=0.95)

# exp() takes them back to vehicles per day
exp(predict(mlr_log2, con, interval='confidence', level=0.95))
exp(predict(mlr_log2, con, interval='prediction', level=0.95))
```